

+ Chapter 5

Internal Memory

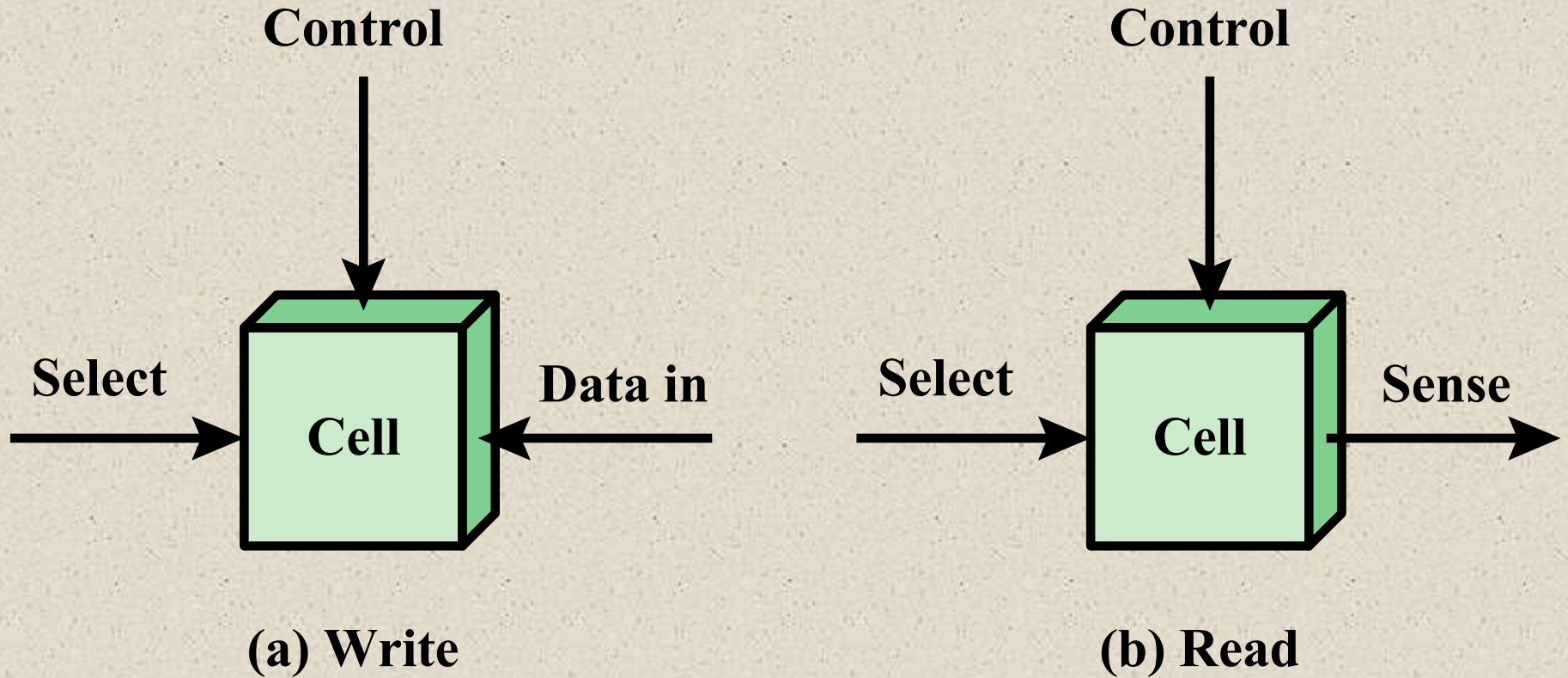
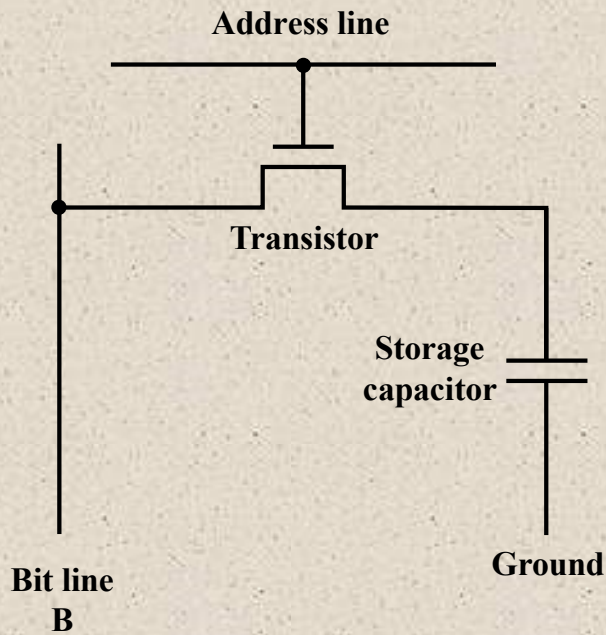


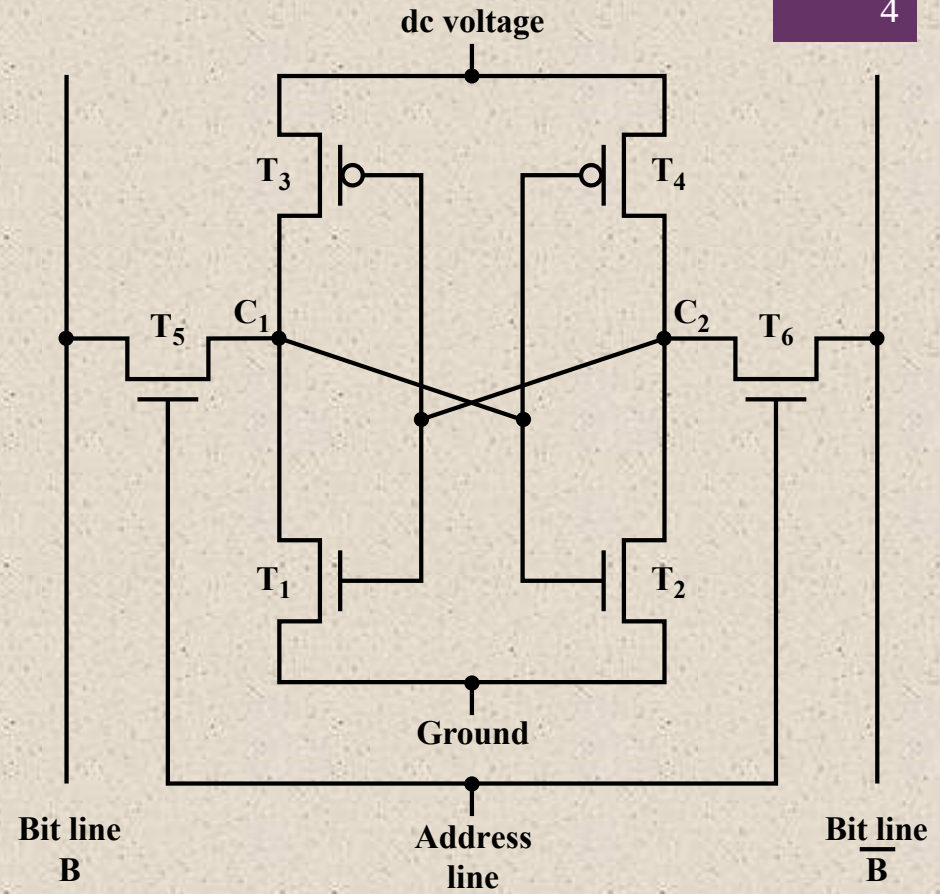
Figure 5.1 Memory Cell Operation

Dynamic RAM (DRAM)

- RAM technology is divided into two technologies:
 - Dynamic RAM (DRAM)
 - Static RAM (SRAM)
- DRAM
 - Made with cells that store data as charge on capacitors
 - Presence or absence of charge in a capacitor is interpreted as a binary 1 or 0
 - Requires periodic charge refreshing to maintain data storage
 - The term *dynamic* refers to tendency of the stored charge to leak away, even with power continuously applied



(a) Dynamic RAM (DRAM) cell



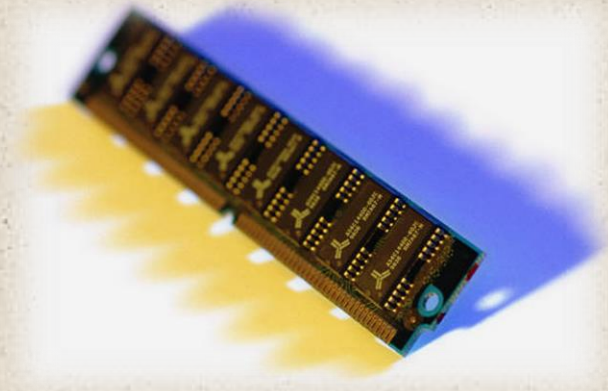
(b) Static RAM (SRAM) cell

Figure 5.2 Typical Memory Cell Structures



Static RAM (SRAM)

- Digital device that uses the same logic elements used in the processor
- Binary values are stored using traditional flip-flop logic gate configurations
- Will hold its data as long as power is supplied to it



SRAM versus DRAM

- Both volatile
 - Power must be continuously supplied to the memory to preserve the bit values
- Dynamic cell
 - Simpler to build, smaller
 - More dense (smaller cells = more cells per unit area)
 - Less expensive
 - Requires the supporting refresh circuitry
 - Tend to be favored for large memory requirements
 - Used for main memory
- Static
 - Faster
 - Used for cache memory (both on and off chip)

SRAM

DRAM

Read Only Memory (ROM)

- Contains a permanent pattern of data that cannot be changed or added to
- No power source is required to maintain the bit values in memory
- Data or program is permanently in main memory and never needs to be loaded from a secondary storage device
- Data is actually wired into the chip as part of the fabrication process
 - Disadvantages of this:
 - No room for error, if one bit is wrong the whole batch of ROMs must be thrown out
 - Data insertion step includes a relatively large fixed cost

Error Correction

■ Hard Failure

- Permanent physical defect
- Memory cell or cells affected cannot reliably store data but become stuck at 0 or 1 or switch erratically between 0 and 1
- Can be caused by:
 - Harsh environmental abuse
 - Manufacturing defects
 - Wear

■ Soft Error

- Random, non-destructive event that alters the contents of one or more memory cells
- No permanent damage to memory
- Can be caused by:
 - Power supply problems
 - Alpha particles

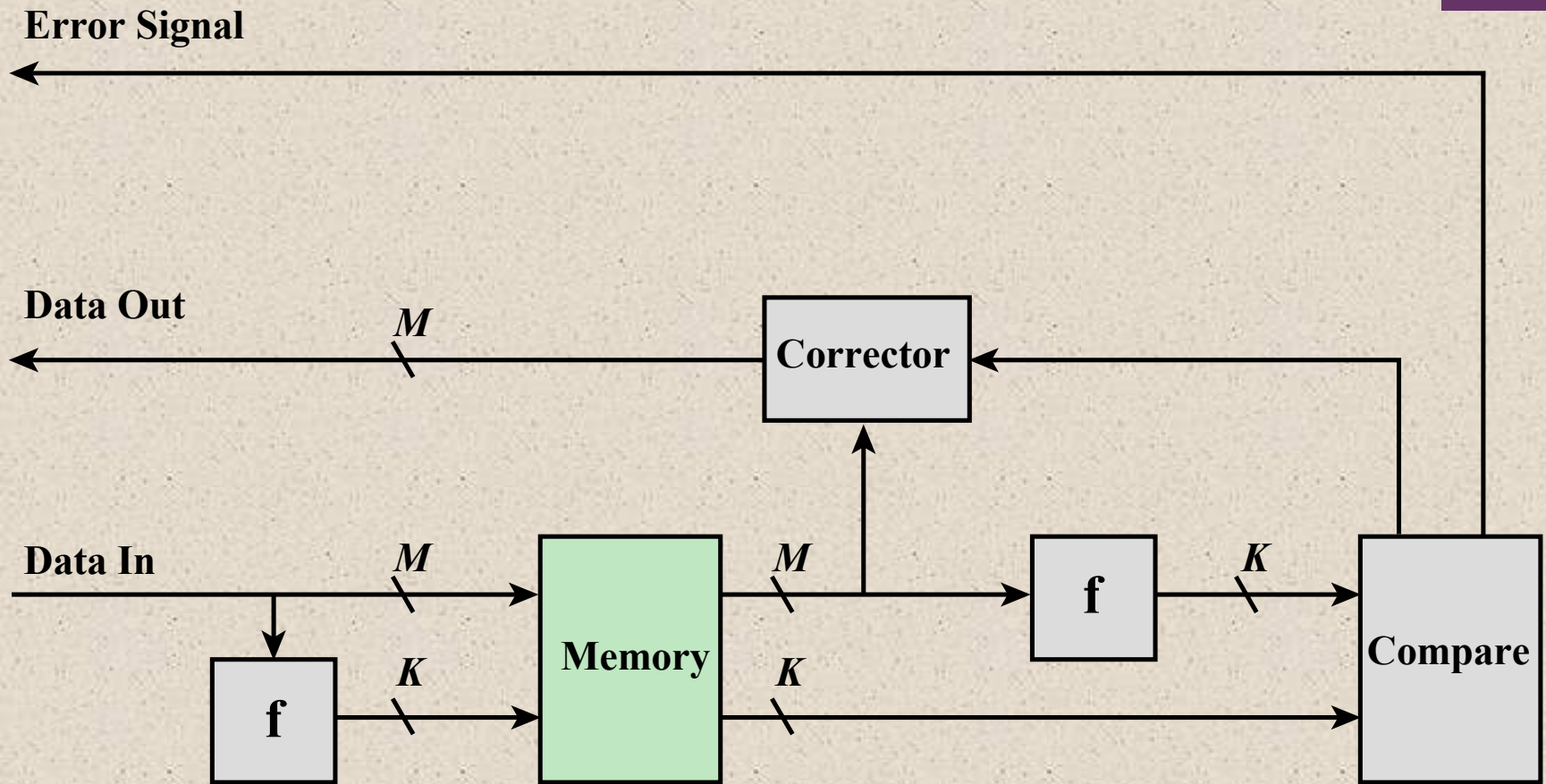


Figure 5.7 Error-Correcting Code Function

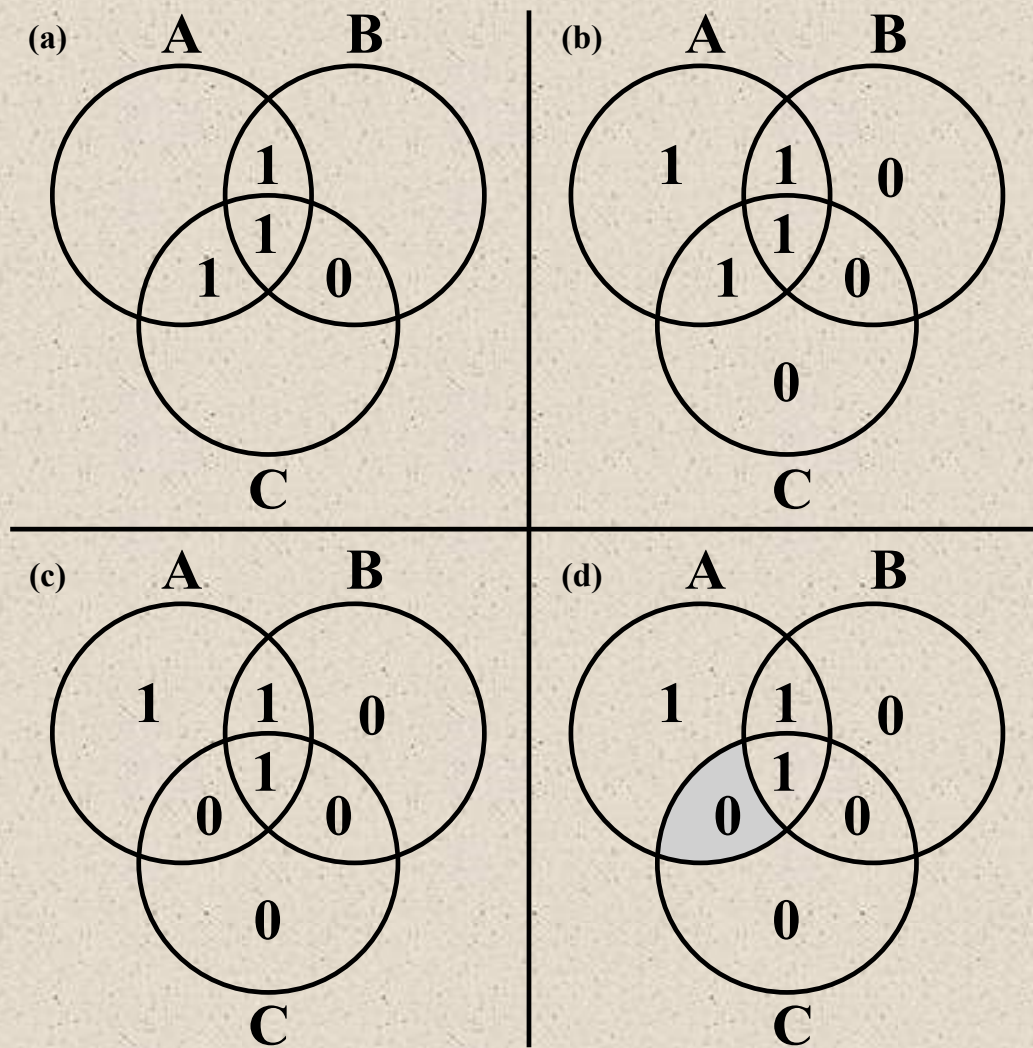


Figure 5.8 Hamming Error-Correcting Code

Bit Position	12	11	10	9	8	7	6	5	4	3	2	1
Position Number	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
Data Bit	D8	D7	D6	D5		D4	D3	D2		D1		
Check Bit					C8				C4		C2	C1

Figure 5.9 Layout of Data Bits and Check Bits

Bit position	12	11	10	9	8	7	6	5	4	3	2	1
Position number	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
Data bit	D8	D7	D6	D5		D4	D3	D2		D1		
Check bit					C8				C4		C2	C1
Word stored as	0	0	1	1	0	1	0	0	1	1	1	1
Word fetched as	0	0	1	1	0	1	1	0	1	1	1	1
Position Number	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
Check Bit					0				0		0	1

Figure 5.10 Check Bit Calculation

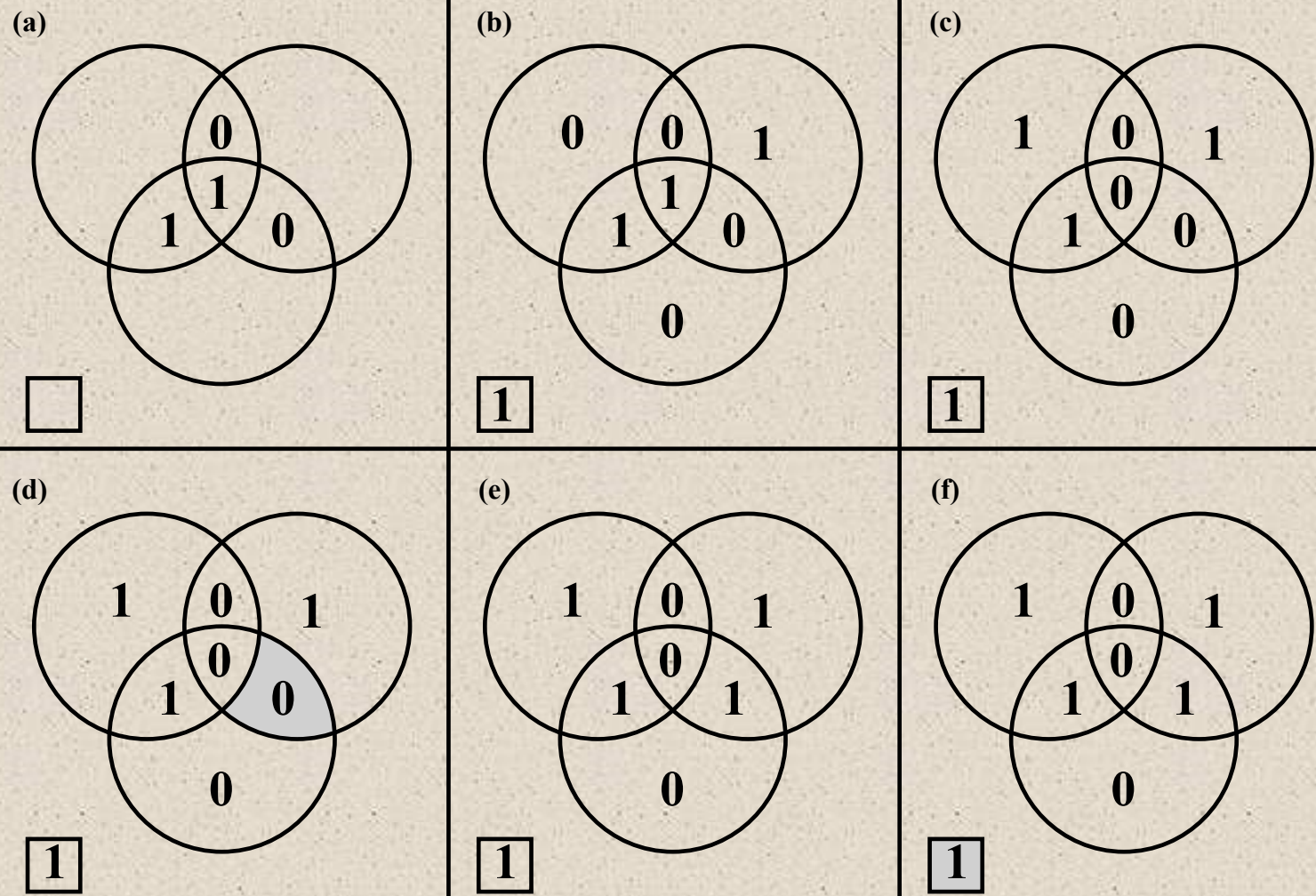


Figure 5.11 Hamming SEC-DED Code



Double Data Rate SDRAM (DDR SDRAM)

- Developed by the JEDEC Solid State Technology Association (Electronic Industries Alliance's semiconductor-engineering-standardization body)
- Numerous companies make DDR chips, which are widely used in desktop computers and servers
- DDR achieves higher data rates in three ways:
 - First, the data transfer is synchronized to both the rising and falling edge of the clock, rather than just the rising edge
 - Second, DDR uses higher clock rate on the bus to increase the transfer rate
 - Third, a buffering scheme is used

	DDR1	DDR2	DDR3	DDR4
Prefetch buffer (bits)	2	4	8	8
Voltage level (V)	2.5	1.8	1.5	1.2
Front side bus data rates (Mbps)	200—400	400—1066	800—2133	2133—4266

Table 5.4
DDR Characteristics

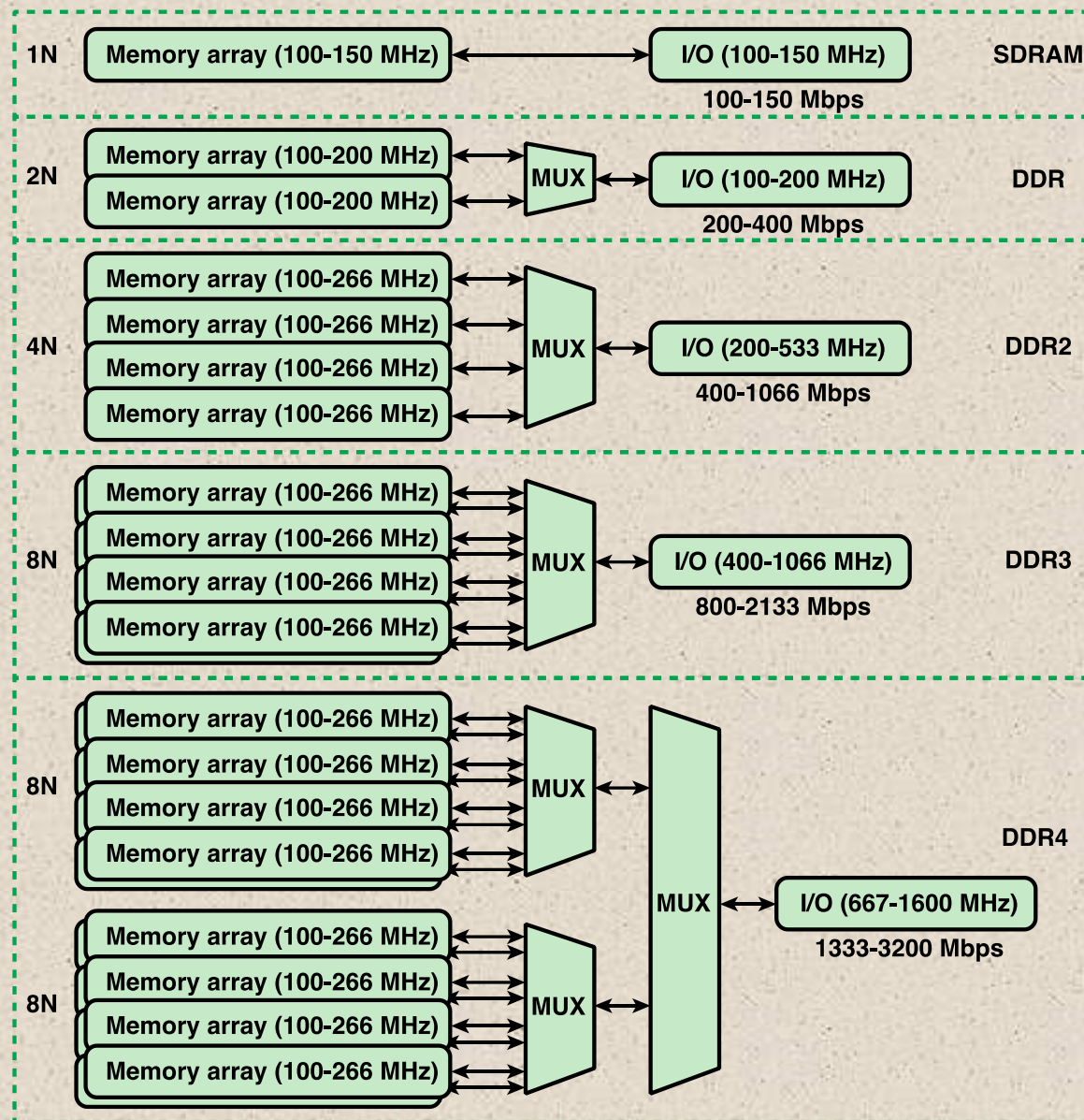
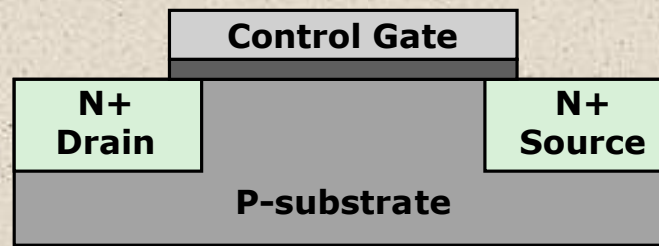


Figure 5.14 DDR Generations

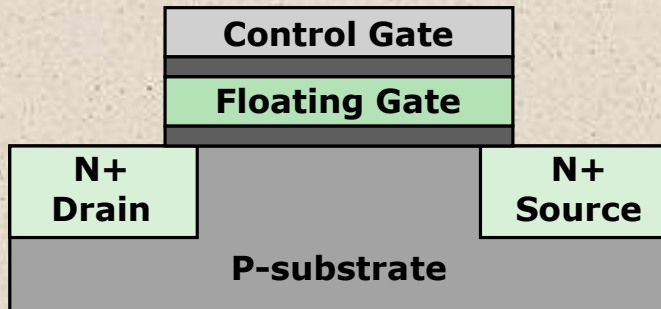


Flash Memory

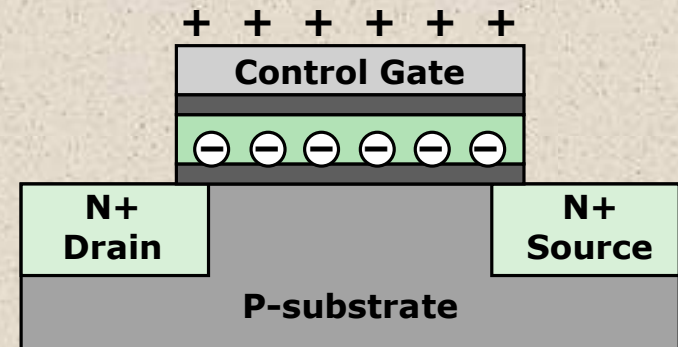
- Used both for internal memory and external memory applications
- First introduced in the mid-1980's
- Is intermediate between EPROM and EEPROM in both cost and functionality
- Uses an electrical erasing technology like EEPROM
- It is possible to erase just blocks of memory rather than an entire chip
- Gets its name because the microchip is organized so that a section of memory cells are erased in a single action
- Does not provide byte-level erasure
- Uses only one transistor per bit so it achieves the high density of EPROM



(a) Transistor structure



(b) Flash memory cell in one state



(c) Flash memory cell in zero state

Figure 5.15 Flash Memory Operation

+ Summary

Chapter 5

Internal Memory

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- Semiconductor main memory
 - Organization
 - DRAM and SRAM
 - Types of ROM
- Error correction
- DDR DRAM
 - Synchronous DRAM
 - DDR SDRAM
- Flash memory
 - Operation
 - NOR and NAND flash memory